Bihar Engineering University, Patna

B.Tech. 3rd Semester Examination, 2023

Course: B.Tech. | *Time: 03 Hours*
Code: 104302 | *Subject: Electrical and Electronic Material* | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 ***Choose the correct answer of the following (Any seven question only):*** **[2 x 7 = 14]**

(a) Which of the following is regarded as the 'repeatable entity' of a 3D crystal structure?
(i) Unit cell (ii) Lattice
(iii) Crystal (iv) Bravais Index

(b) In a silicon crystal, the arrangements of atoms are repeated periodically. This type of material is classified as
(i) Amorphous and non-crystalline (ii) Non-crystalline and epitaxial
(iii) Epitaxial and single crystal (iv) Amorphous and single crystal

(c) An electronic band is
(i) any range of energy levels
(ii) energy levels where free electrons exist
(iii) a range of energy levels characterized by the existence of electrons
(iv) all of these

(d) Which of the following is a semi-conductor
(i) Diamond (ii) Arsenic
(iii) Phosphorous (iv) Gallium arsenide

(e) The movement of a hole results from
(i) Excitation due to high temperature
(ii) The vacancy filled by a valence electron from the neighbouring atom
(iii) Change in number of protons in the atom
(iv) None of above

(f) The magnetization of a superconductor is
(i) 0 (ii) -B
(iii) -1 (iv) -H

(g) Resistance of a super conductor is
(i) infinity (ii) zero
(iii) unity (iv) greater than 1

(h) How can a dielectric be converted to a conductor?
(i) Compression (ii) Expanding
(iii) Heating (iv) Freezing

(i) Basic source of magnetism is
(i) charged particles alone (ii) movement of charged particles
(iii) magnetic dipoles (iv) magnetic domains

(j) The relative permeability is less than 1 in
(i) diamagnetic materials (ii) diamagnetic materials
(iii) paramagnetic materials (iv) ferrites

Q.2 (a) Explain BCS theory for superconductivity. **[7]**
(b) How is mirror finished wafer prepared? **[7]**

Q.3 (a) How the structures of simple crystals are determined by X-ray diffraction? **[7]**
(b) Write a short note on "Reduced E-K diagram". **[7]**

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Q.4 (a) Discuss the factors affecting polarization. [6]

(b) What are degenerate and non-degenerate semiconductors? Explain with suitable energy band diagram. [8]

Q.5 (a) Discuss in detail about the packing geometry in metallic, ionic and covalent solids. [10]

(b) What are single and double-sided PCBs? [4]

Q.6 (a) Define Hall coefficient and Hall mobility. Mention some application area of Hall effect. [7]

(b) An Si sample has been doped with 10^{17} arsenic atoms/cm^{-3}. Calculate the conductivity of the sample at 127 °C (400 K). The above n-type Si sample is further doped with 9×10^{16} boron atoms/cm^{-3}. Calculate the conductivity of the sample at 127 °C. [7]

Q.7 (a) Discuss the influence of temperature on magnetic behaviour of a material. [6]

(b) Define magnetic hysteresis using hysteresis loop. Also discuss about ideal and non-ideal magnetic material. [8]

Q.8 Differentiate the following: [14]

(a) Crystal lattice and unit cell

(b) amorphous and crystalline solids

(c) N-type and p-type semiconductors

(d) Intrinsic and extrinsic semiconductor

Q.9 Write short notes on *any two* of the following: [7x2=14]

(a) Manufacturing techniques of PCBs

(b) Importance of PCB in electrical and electronics engineering

(c) Piezoelectricity

(d) Dielectric losses

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